Pharmacology of DMARDS Part 1:

Disease Modifying Antirheumatic Drugs

Pharmacology of Glucocorticoids and Methotrexate

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Following completion of the preparation materials, students should be able to:

1. List the clinically relevant pharmacokinetic properties of the glucocorticoids and each of the disease modifying antirheumatic drugs (DMARDS) starred (✳) on the notes handout.
2. Explain the mechanisms of action and mechanisms of adverse effects of the glucocorticoids, each of the DMARDs classes, and the starred (✳) agents handout using specific examples.
3. Describe the roles of NSAIDs, glucocorticoids, and each class of DMARDs in the treatment of rheumatoid arthritis in relation to their mechanisms and effects using specific examples from among the starred (✳) agents.
4. Relate the toxicities of the glucocorticoids and each of the DMARDs to their effects in the various tissues using specific examples.
5. Identify the pretreatment testing and monitoring recommended during therapy in relation to potential adverse effects of DMARDs therapy.
6. Develop a visual aid (concept map, table, or whatever works for you) of the glucocorticoids' and DMARDs' pharmacological effects and risk factors correlated with their potential drug-disease and drug-drug interactions and precautions in special populations (pregnancy and lactation, pediatric and geriatric patients).

Preparation Materials

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the CPG and on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links to resources

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ScholarRx Brick > Immunology

- Autoimmune Diseases and Autoinflammatory Disease: https://exchange.scholarrx.com/collection/immunology
- Treatment of Immune Disorders: https://exchange.scholarrx.com/brick/rheumatoid-arthritis
- Biologic Agents: https://exchange.scholarrx.com/brick/biologic-agents

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36: Nonsteroidal Anti-inflammatory Drugs, Disease Modifying Antirheumatic Drugs, Nonopioid Analgesics & Drugs Used in Gout

The section on Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281752931

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 39: Adrenocorticosteroids & Adrenocortical Antagonists > The Naturally Occurring Glucocorticoids; Cortisol (hydrocortisone) – The sections on Naturally Occurring Glucocorticoids and Synthetic Corticosteroids > Adrenocorticosteroids

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281753522#1204142345

LWW Heath Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 40: Anti-inflammatory, Antipyretic, and Analgesic Agents

https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253330080&bookid=3222

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 36: NSAIDs, Acetaminophen & Drugs Used in Rheumatoid Arthritis & Gout

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285596163

Introduction: Pathophysiologic basis of DMARDs therapy.

- Inappropriate activation of the immune system can result in inflammation and immune-mediated diseases such as rheumatoid arthritis (RA). Normally, the immune system can differentiate between self and nonself.
- In RA, white blood cells (WBCs) recognize the synovium as nonself and initiate an inflammatory attack. WBC activation leads to stimulation of T lymphocytes, which recruit and activate monocytes and macrophages. These cells secrete proinflammatory cytokines, including tumor necrosis factor-α (TNF), IL-6, other interleukins, and growth factors into the synovial cavity, ultimately leading to joint destruction and systemic abnormalities characteristic of RA.
- B lymphocytes produce rheumatoid factor and other autoantibodies that maintain inflammation with progressive tissue injury resulting in joint damage and erosions, functional disability, pain, and reduced quality of life.
- Pharmacotherapy for RA includes immunosuppressive agents (DMARDs) that reduce the inflammatory process. The goals of RA treatment are tight control of disease activity and, ideally, remission.

What you need to know and understand

- Much of the joint damage that results in disability begins early in the course of the disease. Better outcomes are achieved when DMARD therapy is begun early in RA patients. Disease activity should be assessed frequently.
- Recommended initial treatment for active RA is DMARD monotherapy or combination therapy, along with rapid-acting anti-inflammatory drugs – glucocorticoids or NSAIDs – as "bridging therapy", because onset of the DMARDs' antirheumatic effects is delayed from weeks to months. Factors affecting the choice of drugs are the level of disease activity, comorbid conditions, stage of therapy (initial or subsequent), insurance coverage, patient preferences, and signs of adverse effects.
- The three subclassifications of DMARDs are conventional synthetic DMARDs (csDMARDs), biologic DMARDs (bDMARDs), and small molecule targeted synthetic agents (tsDMARDs).
- Patients must be screened before treatment for malignancy and the presence of active or latent infections – tuberculosis, hepatitis B and hepatitis C – as well as comorbidities, and monitored for infections and adverse drug effects during therapy.

What you need to know and understand about csDMARDs

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- The currently used csDMARDs are methotrexate (MTX), sulfasalazine (SSZ), hydroxychloroquine (HCQ), and leflunomide (LEF). All are taken orally for RA.
- MTX is a folic acid antagonist that inhibits purine nucleotide biosynthesis and cytokine production, leading to immunosuppression. It is the usual initial therapy and the mainstay (the "anchor") of RA treatment. It is frequently used in combination with the other DMARDs from all subclasses. For RA, MTX is taken in low, once weekly doses, with folic acid supplementation.
- MTX is taken up by cells via reduced folate transporter. Glutamate residues are added to the molecule intracellularly (polyglutamation → MTX-PG), which accumulates inside the cell. MTX and MTX-PG block dihydrofolate reductase (DHFR), thymidylate synthase, and an enzyme called ATIC, which is important in the generation of inosine monophosphate and the metabolism of AMP and GMP. MTX's immunosuppresive effect is thought to be due to the extracellular actions of adenosine. (See lecture for details.)
- MTX is metabolized intracellularly and excreted in urine as unchanged drug. It has a narrow therapeutic window. Common side effects are mucositis, cytopenias, hepatotoxicity, and pulmonary toxicity. MTX is fetotoxic and contraindicated in pregnancy.

What you need to know and understand about csDMARDs

- SSZ is converted by gut microbial flora to its components, sulfapyridine and 5-aminosalicylic acid (mesalamine). Sulfapyridine is absorbed but mesalamine is not. SSZ’s mechanism is unknown. It is frequently used as initial therapy in combination with MTX. Sulfapyridine is metabolized by acetylation (NAT2). Metabolites are excreted in urine. SSZ can cause GI side effects, headache, folic acid deficiency, leukopenia, and reversible oligospermia. SSZ is considered safe in pregnancy.
- HCQ is used for early mild RA. Onset of HCQ’s anti-RA effects is 6 weeks to 6 months. It is often in triple therapy with MTX and SSZ for enhanced efficacy. The mechanism is unclear but may be related to altering the pH of lysosomes, thus interfering with lysosome function. HCQ has a very large volume of distribution, as it sequesters in liver, spleen, and melanin-containing tissues. It can cause ocular toxicity (retinopathy and corneal deposits), GI upset, and CNS effects. It deposits in melanocytes and causes skin discoloration and eruptions. It is considered safe in pregnancy.
- Leflunomide inhibits dihydroorotate dehydrogenase, disrupting pyrimidine synthesis, causing cell cycle arrest (G1/S phase) in autoimmune lymphocytes. It may be used as monotherapy in patients who are intolerant to MTX. LEF is a prodrug converted to the active metabolite, teriflunomide. It undergoes enterohepatic circulation and has a very long half-life. Common adverse effects are headache, diarrhea, nausea, hepatotoxicity, cytopenias, pulmonary toxicity, and peripheral neuropathy. Leflunomide is teratogenic/fetotoxic and contraindicated in pregnancy.

What you need to know and understand about the DMARDs class

- bDMARDs are monoclonal antibodies or antibody-like proteins (fusion proteins) produced through recombinant technology and administered intravenously or subcutaneously. They target cytokines, cytokine receptors, or other cell surface molecules. The bDMARDs subclasses are the anti-TNF agents, the IL-6 receptor antagonists, a T cell coreceptor blocker that prevents T cell activation, and a CD20+ B cell depleting agent. Infusion and hypersensitivity reactions may occur with the proteinaceous drugs.
- The TNF inhibitors (anti-TNF agents) are infliximab, adalimumab, etanercept, golimumab, and certolizumab pegol. They have structural differences but similar efficacy. Their mechanism of action is binding to soluble and membrane bound TNF, which prevents this cytokine from activating its receptors. Etanercept binds only soluble TNF. Clinical response can be seen in 2 weeks. A TNF inhibitor may be a preferred initial therapy in patients with moderately to severely active disease. They are effective as monotherapy and may be used with MTX and other csDMARDs. Infliximab is administered intravenously. The others are by subcutaneous injection.
- Anti-TNF agents can cause or worsen heart failure, induce reactivation of tuberculosis and viral hepatitis, and increase the risk of lymphoma and other malignancies. They should not be used with other biologic immunosuppressants or small molecule targeted agents because of the risk of severe infection. If used during pregnancy, the general expert recommendation is to discontinue drug administration at week 32.

What you need to know and understand about bDMARDs

- Tocilizumab and sarilumab are recombinant monoclonal antibodies. They are competitive inhibitors at IL-6 receptors, which prevents activation of the IL-6 signaling pathway. IL-6 activates T cells, B cells, macrophages, and osteoclasts and mediates hepatic acute phase response, and other effects. IL-6 receptor blockers are used for the treatment of RA and other inflammatory diseases. Tocilizumab is administered intravenously or subcutaneously for RA. Sarilumab is a subQ agent. They are generally well tolerated. Common side effects in clinical trials were upper respiratory infection, nasopharyngitis, headache, and elevated blood pressure, liver enzymes, and cholesterol.
- bDMARDs that exert immunosuppression by acting on lymphocytes are abatacept (T cell costimulation inhibitor) and rituximab (B cell depleting agent).
- Abatacept is a fusion protein (CTLA-4 fused to IgG) that inhibits T cell activation by binding to CD80/86 (B7.1/B7.2) on the antigen presenting cell, which blocks the costimulatory interaction with CD28 on the T cell. Abatacept is indicated for patients with moderately to severely active disease. Abatacept is administered I.V. or subQ for RA. Safety and pretreatment and monitoring recommendations are comparable to the TNF inhibitors.

What you need to know and understand about bDMARDs/tsDMARDs

- Rituximab eliminates CD20-positive B cells through complement-mediated cytotoxicity, antibody-dependent cellular toxicity, and stimulation of apoptosis. It was developed for the treatment of CD20+ B cell lymphomas and leukemia and has application for several autoimmune diseases. Infusion reactions, neutropenia, severe mucocutaneous reactions and HBV reactivation, among others are potential complications of rituximab therapy. Rituximab has been associated with case reports of progressive multifocal leukoencephalopathy (PML), which is a neurological disorder characterized by destruction of the cells that produce the myelin sheath in the brain in spinal cord. It is caused by JC virus. Patients receiving rituximab should be monitored for symptoms/signs of PML. Rituximab is given I.V. for RA.
- The tsDMARDs, tofacitinib, baracitinib, and upadacitinib, are oral JAK inhibitors. They act at the cytoplasmic region of transmembrane cytokine receptors and prevent phosphorylation by Janus kinases (JAKs) (protein tyrosine kinases). JAKs activate STATs, which transduce cellular signaling. Cellular responses include proliferation, differentiation, migration, apoptosis, and cell survival. JAK inhibitors reduce the signs/symptoms of disease activity in patients with RA. JAK inhibitors are associated with increased risk of serious infection, heart attack, stroke, cancer, thrombosis, and death.

What you need to know and understand about glucocorticoids (GCs)

Glucocorticoids are highly effective at relieving inflammation associated with RA.

1. Synthetic GCs mimic the actions of cortisol.
2. GCs act via several pathways, including inhibition of the transcription factors, nuclear factor-κB (nf-κB). They decrease the synthesis of proinflammatory cytokines and enzymes, decrease T cell function, and reduce Fc receptor (antibody receptor) expression. Glucocorticoids block all known pathways of eicosanoid synthesis.
3. Short-term low-dose glucocorticoids in RA decreases clinical disease activity. These agents have a beneficial effect on radiographic activity.
4. Myriad acute and delayed adverse events are common with these drugs. Many of the side effects are dose-related. *Careful titration of dose and strategies to minimize toxicity are essential when glucocorticoids are used*.
5. Administration of pharmacologic glucocorticoids causes suppression of the hypothalamic-pituitary-adrenal (HPA) axis, which leads to adrenal insufficiency. Abrupt cessation or too rapid withdrawal of glucocorticoid therapy can cause symptoms of adrenal insufficiency.
6. GCs are administered orally or intra-articularly for RA and intravenously for severe inflammation. They are well absorbed from the GI tract, highly plasma protein bound, and widely distributed. They are extensively metabolized by the liver. Metabolites and excreted in urine. Half-lives vary.

Pause the video and meet Mary, a 37-year-old woman who presents to the rheumatologist with a 6-month history of joint pain, stiffness, and swelling primarily affecting her hands and wrist.

- Mary reports that symptoms are worse in the morning, lasting about an hour. The symptoms began gradually and have progressively worsened over the last 6 months. She has felt fatigued for the past month.
- She reports, difficulty dressing, opening doors, and working at the computer, which is affecting her ability to do her job. Ibuprofen 400 mg in the morning helps ease the pain but the fingers and wrist still seem stiff and swollen. When asked, Mary states that she has a contraceptive implant and is not planning on becoming pregnant.
- “Can you help, doctor?”
- Mary has no known drug allergies (NKDA) and immunizations are up to date.
- Visible swelling, limited range of motion, and warmth and redness are noted in the MCP and PIP joints.
- Plain radiographs and serology confirm the diagnosis of RA. Testing for latent TB, HBV, and HCV are performed.
- Mary is given information about RA and its prognosis and counseled on nutrition, exercise, and physical and occupational therapy.

    What is the goal of RA pharmacologic therapy?

    What is the role of NSAIDs and glucocorticoids in RA management?

    What drug is the DMARD of choice for the initial treatment of active RA?

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RA is a chronic, systemic inflammatory disorder

Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

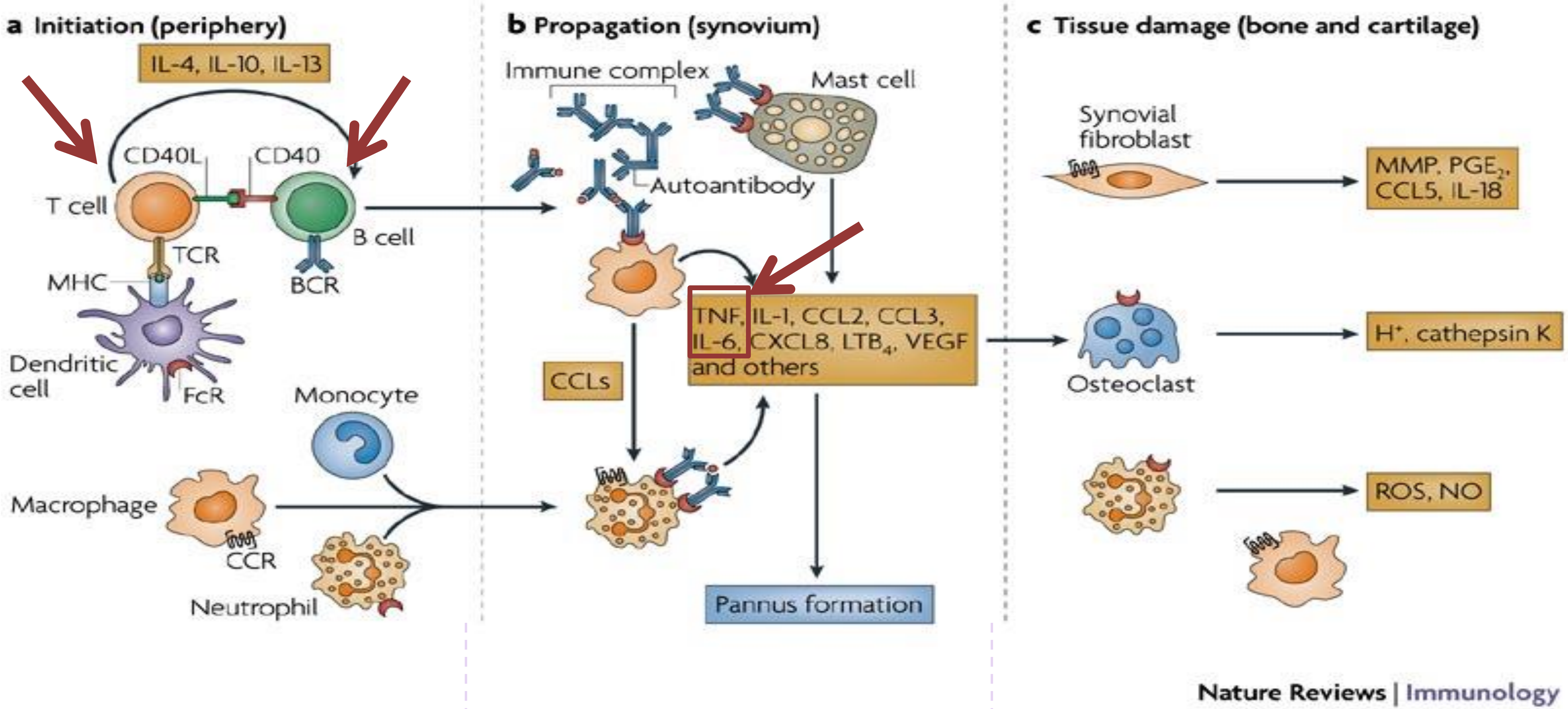
- Joint disease that reduces mobility and quality of life
- Extraarticular manifestations include
  - fatigue
  - subcutaneous nodules
  - lung involvement
  - pericarditis
  - peripheral neuropathy
  - vasculitis
  - hematologic abnormalities

Lower figure: Harrison's Principles of Internal Medicine, 20e; figure 351-1: Metacarpophalangeal and proximal interphalangeal joint swelling in rheumatoid arthritis.

The Goals of RA therapy

1. **early, aggressive pharmacotherapy** to prevent joint damage and disability
2. **modification** of therapy with utilization of combination therapy where appropriate
3. **individualization** of therapy in an attempt to maximize response and minimize side effects
4. **remission** of clinical disease activity, whenever possible

Arrows point to drug targets.



Antigen presentation can occur in peripheral lymphoid organs and in the synovium

Immune complexes binding to Fc receptors on macrophages, neutrophils and mast cells → activation → release of pro-inflammatory cytokines and chemokines → leukocyte infiltration and pannus formation

Activation and recruitment of synovial cells, which release inflammatory mediators → synoviocytes invade cartilage → irreversible bone and cartilage destruction

stepwise progression of the development of rheumatoid arthritis

http://www.nature.com/nri/journal/v7/n3/fig_tab/nri2036_F2.html March 2007

Pharmacologic agents used in the management of RA:

NSAIDs for symptomatic relief

Glucocorticoids for rapid anti-inflammatory action

csDMARDs

bDMARDs

tsDMARDs

csDMARDs: conventional synthetic DMARDs
bDMARDs: biologic DMARDs
tsDMARD: small molecule targeted synthetic DMARD

RA Therapy options

csDMARDs	bDMARDS	
✳Methotrexate ✳Hydroxychloroquine ✳Sulfasalazine ✳Leflunomide	TNFα Inhibitors	✳Infliximab ✳Adalimumab ✳Etanercept Golimumab Certolizumab pegol
tsDMARDs	**IL-6R Inhibitors:**	**Lymphocyte modulators:**
✳Toficitinib Baricitinib Upadacitinib	✳Tocilizumab Sarilumab	✳Abatacept (T cell) ✳Rituximab (B cell)

Bridging therapy for rapid reduction of inflammation and relief of symptoms before the onset of fully effective DMARD therapy

NSAIDs

NSAIDs have minimal, if any, effect on the progression of bone and cartilage destruction but are useful in managing pain and inflammation.

Glucocorticoids, low- to moderate-dosing

Glucocorticoids are powerful anti-inflammatory agents that are highly effective at relieving the inflammation associated with RA.

- Bridging therapy
- Management of acute disease flares
- Intra-articular for transient symptomatic relief when systemic medical therapy has failed to resolve inflammation.

General approach to pharmacological treatment of RA
Expert recommendation: Early use of DMARDS

- ➢ All patients diagnosed with RA should be started on DMARD therapy as soon as possible.

The choice of drug therapy depends on the:

- Level of disease activity
- Presence of comorbid conditions
- Stage of therapy (initial or subsequent therapy)
- Insurance company limitations
- Patient preferences
- Presence of signs associated with poorer outcome

References are listed at UpToDate: General principles and overview of management of rheumatoid arthritis in adults

Early RA goal of therapy: Clinical remission

Aggressive treatment with DMARDs can lead to better outcomes

Initial therapy:

- Either monotherapy with escalation or DMARD combination
- Early RA and features of poor prognosis warrants aggressive therapy

Methotrexate is preferred by most experts as monotherapy and the "anchor" in combinations with csDMARDs, bDMARDs, tsDMARDs.

Improved clinical outcomes were found with combination of MTX and most bDMARDs compared to either drug alone.

Patients who have had an inadequate response to anti-TNF agents:

- MTX combination therapy with csDMARD, bDMARD, or tsDMARD

Treatment of RA in Pregnancy

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Recommendation	Drugs	Comments
Drugs with minimal fetal and maternal risk that can generally be used during pregnancy	Hydroxychloroquine	Considered safe
	Sulfasalazine	Considered safe
	Azathioprine (consensus opinion)	Several studies have found no increased risk of congenital abnormalities, low birthweight, early childhood infections, autism spectrum disorder, or attention deficit hyperactivity disorder when exposed in utero to AZA.
	TNF inhibitors	Most professional societies recommend discontinuation at week 32 of gestation. Certolizumab pegol may be used throughout pregnancy.
Drugs potentially safe during pregnancy, with certain limitations	Prednisone Prednisolone	Very low concentrations cross the placenta (use the lowest effective dose).
	NSAIDs Aspirin, higher doses	Safe until 20 weeks gestation, then they should be discontinued due to potential fetal adverse effects. If needed, monitor for oligohydramnios.
Other bDMARDs are not currently recommended for the treatment of rheumatic or musculoskeletal diseases.		
Contraindicated	Methotrexate Leflunomide	Teratogenic and Fetotoxic

Infants, Children, Adolescents, and Older Patients

LACTATION Weigh the risk to the fetus and the benefit to the mother.

(Some agents are considered compatible with breastfeeding.)

PEDIATRICS Dosing information is available for children and adolescents with rheumatic and musculoskeletal diseases.

GERIATRICS Older patients may have comorbid conditions and may be more susceptible to adverse effects and drug interactions.

Early RA therapeutic target: Low disease activity or Remission

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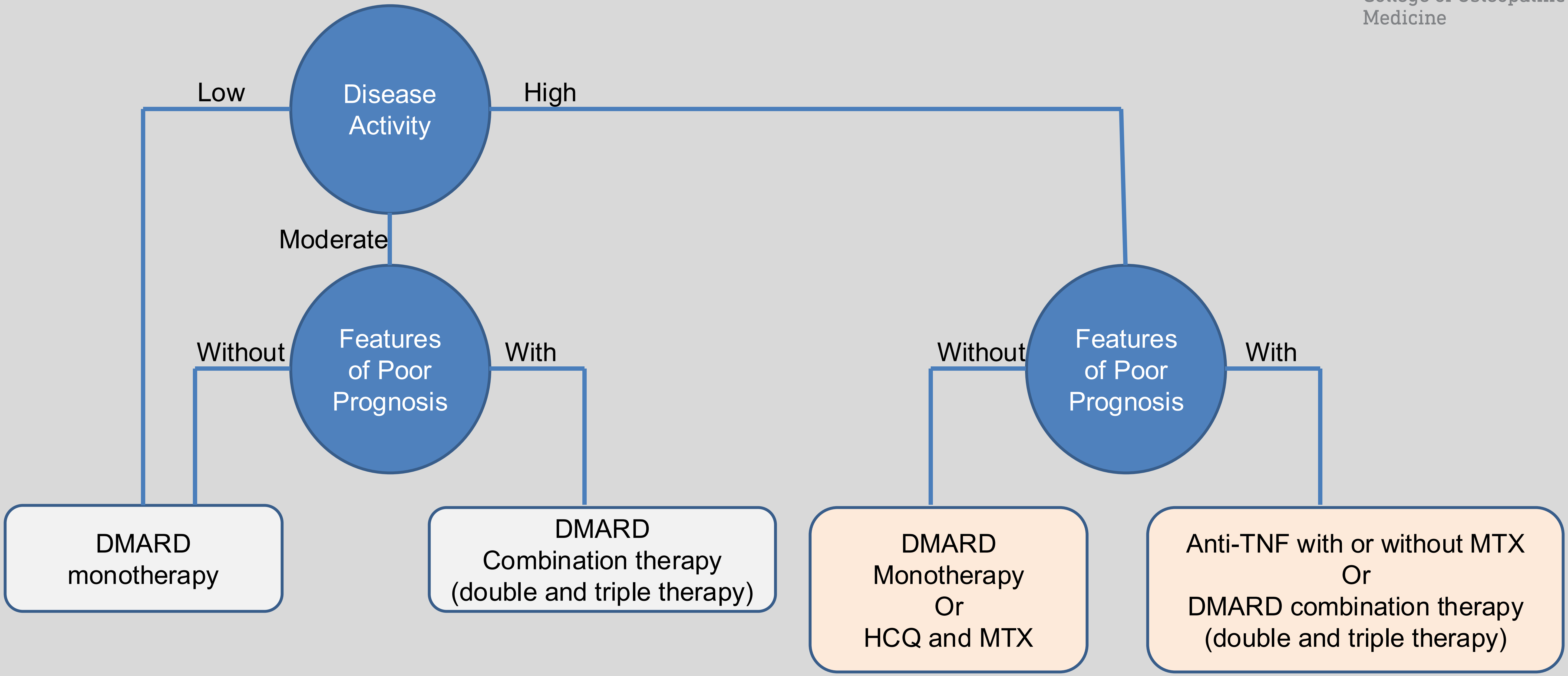
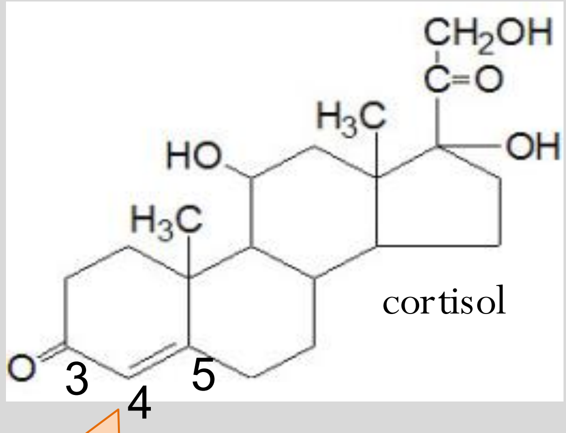


2012 Update of 2008 American College of Rheumatology Recommendations

Glucocorticoids

- Prednisone
- Methylprednisolone

and many others



4,5 double bond on A ring essential for GC function

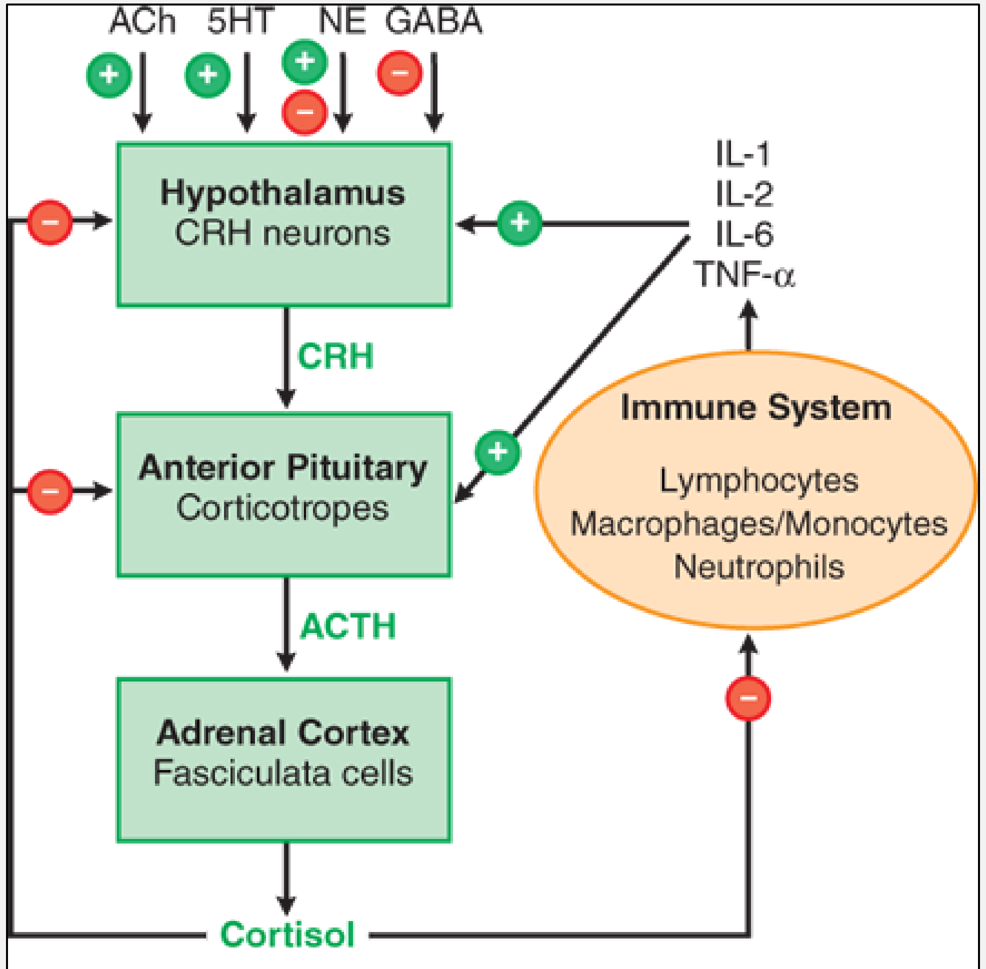
Glucocorticoids Class Pharmacokinetics

A	Lipophilic – well absorbed from GI tract, and also from sites of local administration – oral, parenteral, topical preparations
D	Transport: CBG and Albumin; Saturable
M	Hepatic: Agent-dependent Cortisone (inactive) / Prednisone (inactive) $\xrightarrow{\text{11β-HSD1}}$ Cortisol (active) / Prednisolone (active) Cortisol (active) / Prednisolone (active) $\xrightarrow{\text{11β-HSD2}}$ Cortisone (inactive) / Prednisone (inactive)
E	Renal; glucuronide and sulfate metabolites
t½	varies depending on alterations in GC molecule: biological activity often lasts much longer because of the profound changes that these drugs exert on gene expression profiles within their target cells

CYP3A family is a major pathway in the metabolism of most synthetic GCs

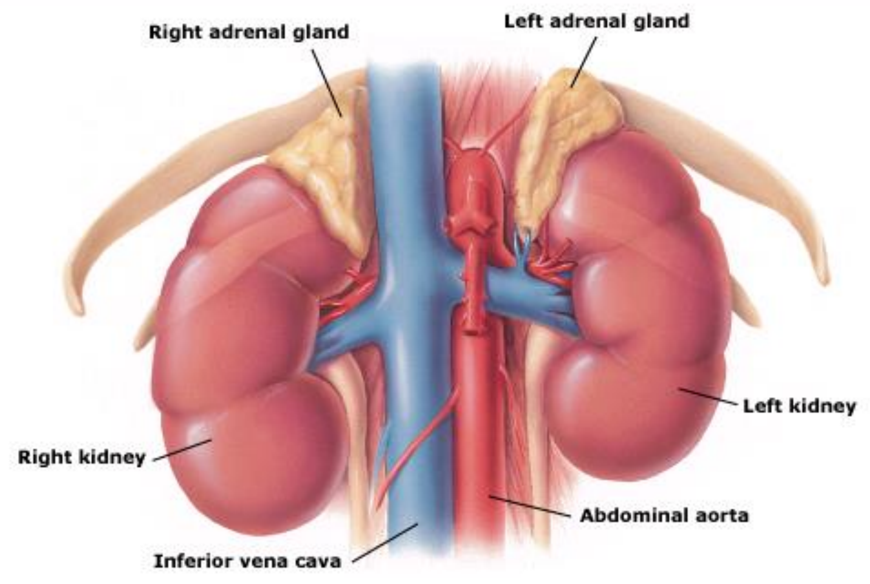
CBG: corticosteroid binding globulin
11β-HSD1: 11β-hydroxysteroid dehydrogenase type 1; 11β-HSD2: 11β-hydroxysteroid dehydrogenase type 2

The hypothalamic-pituitary-adrenal (HPA) axis maintains appropriate levels of cortisol, which is secreted from the adrenal glands.
Prolonged glucocorticoid therapy suppresses the HPA axis.



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition:

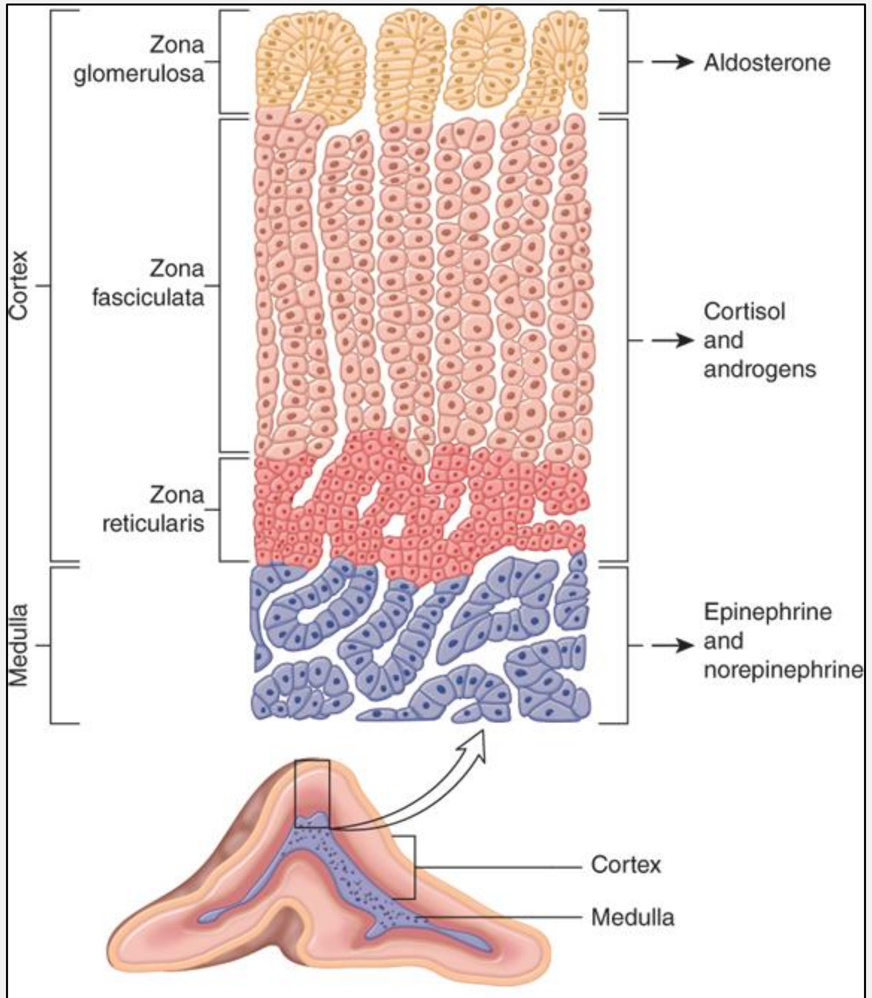
Adrenal glands in relation to the kidneys and great vessels



The adrenal glands are retroperitoneal and are located on the superior medial aspect of the upper pole of each kidney. This illustrates the frontal or intra-abdominal view of the kidneys and adrenal glands.
Reproduced with permission from: Anatomical Chart Company. Copyright © 2008 Lippincott Williams & Wilkins.

UpToDate

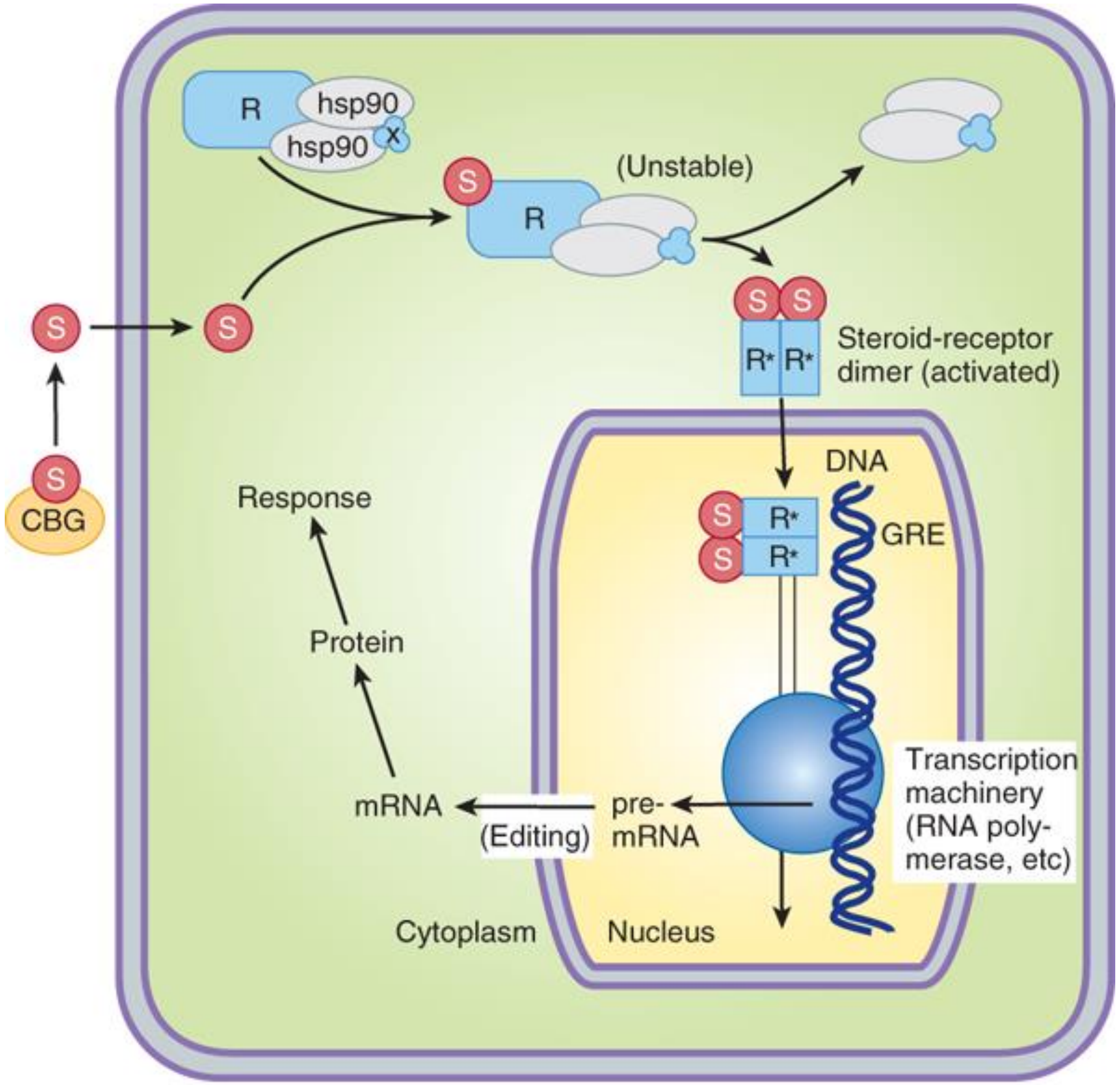
UpToDate Graphic 61177 Version 3.0



Source: David G. Gardner, Dolores Shoback: Greenspan's Basic & Clinical Endocrinology, Tenth Edition

Chapter 46 Adrenocorticotropic Hormone, Adrenal Steroids, and the Adrenal Cortex, Brunton LL, Hilal-Dandan R, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e;* 2017.

Citation: Chapter 9 Glucocorticoids and Adrenal Androgens, Gardner DG, Shoback D. *Greenspan's Basic & Clinical Endocrinology, 10e;* 2017. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=166249274&bookid=2178



Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition

Glucocorticoids Mechanisms of Action

Please see the next slide.

Citation: Chapter 39 Adrenocorticosteroids & Adrenocortical Antagonists, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250600694 Accessed: March 24, 2021

Glucocorticoids Genomic Mechanisms of Action

Cortisol (S) enters cells by passive diffusion
and binds receptor (R)

↓

S-R interaction induces a conformational change in R

↓

R dissociates from chaperone proteins
(HSPs and immunophilin)

↓

S-R complex dimerizes and translocates to the nucleus

↓

S-R complex dimers interact with GRE
which activates transcription
or
complex with other transcription factors (*eg* NFκB, AP-1)
which induces transrepression of genes normally
upregulated by them

Alternate MOAs:

1. S-R complex alters function of other transcription

2. Steroid binding to membrane receptors may cause immediate effects.

Cortisol's effects are broad and numerous.
Synthetic glucocorticoids mimic the actions of cortisol.

- Maintain metabolic homeostasis
  - Carbohydrate, protein, lipid metabolism
  - Electrolyte balance
- Cardiovascular, skeletal muscle, and CNS effects
- Suppress immune system
- Permissive effects on hormones

Glucocorticoids suppress lymphocyte and macrophage ***numbers***, and profoundly alter the immune ***responses*** of lymphocytes and macrophages. The net effect of these actions on various cell types is to markedly diminish the inflammatory response.

Actions of endogenous cortisol and synthetic GCs	**Effect**
Decrease number and activity of inflammatory cells	Reduces inflammatory response
Decrease movement of the cells from the circulation into the area of injury in the tissues	Decreases serum extravasation, swelling, and discomfort.
Prevent neutrophil degranulation	Prevents release of their proinflammatory mediators.
Induce movement of the circulating leukocytes into the lymphoid tissue	Limits the leukocyte activity, which reduces overall inflammatory response in peripheral tissues
Reduce B cell clonal expansion	Limits production of auto-antibodies, which can reduce autoimmune attacks
Reduce T cell proliferation	Impairs activation and function of other immune cells, which can reduce autoimmune attacks
Inhibit gene expression of proinflammatory mediators	Decreases levels of COX-2, NO synthase, cytokines, chemokines, other inflammatory proteins.
Induce gene expression of counterbalancing anti-inflammatory proteins, such as lipocortin-A which suppressed phospholipase A2	Suppression of PLA2 prevents the release of arachidonic acid, which prevents the synthesis of prostaglandins and leukotrienes and other lipoxygenase products

Glucocorticoid Therapy in RA

- **Bridging therapy** (oral)

Dosing based on clinical experience and patient response:

Initiation: Start with a high dose to rapidly decrease inflammation, then taper to a lower maintenance dose – the lowest effective dose

Maintenance: Continue the lower dose GC for symptom control.

Withdrawal: Gradually taper and discontinue the GC when the concomitant DMARD therapy shows effect.

- GC therapy can slow the appearance of new bony erosion.
- GCs are not specific and not curative.

- **Intra-articular therapy** (injection in affected joint)

Symptomatic treatment of one to a few joints

- Management of **acute exacerbation** of RA symptoms
- Control of **extra-articular** manifestations of RA

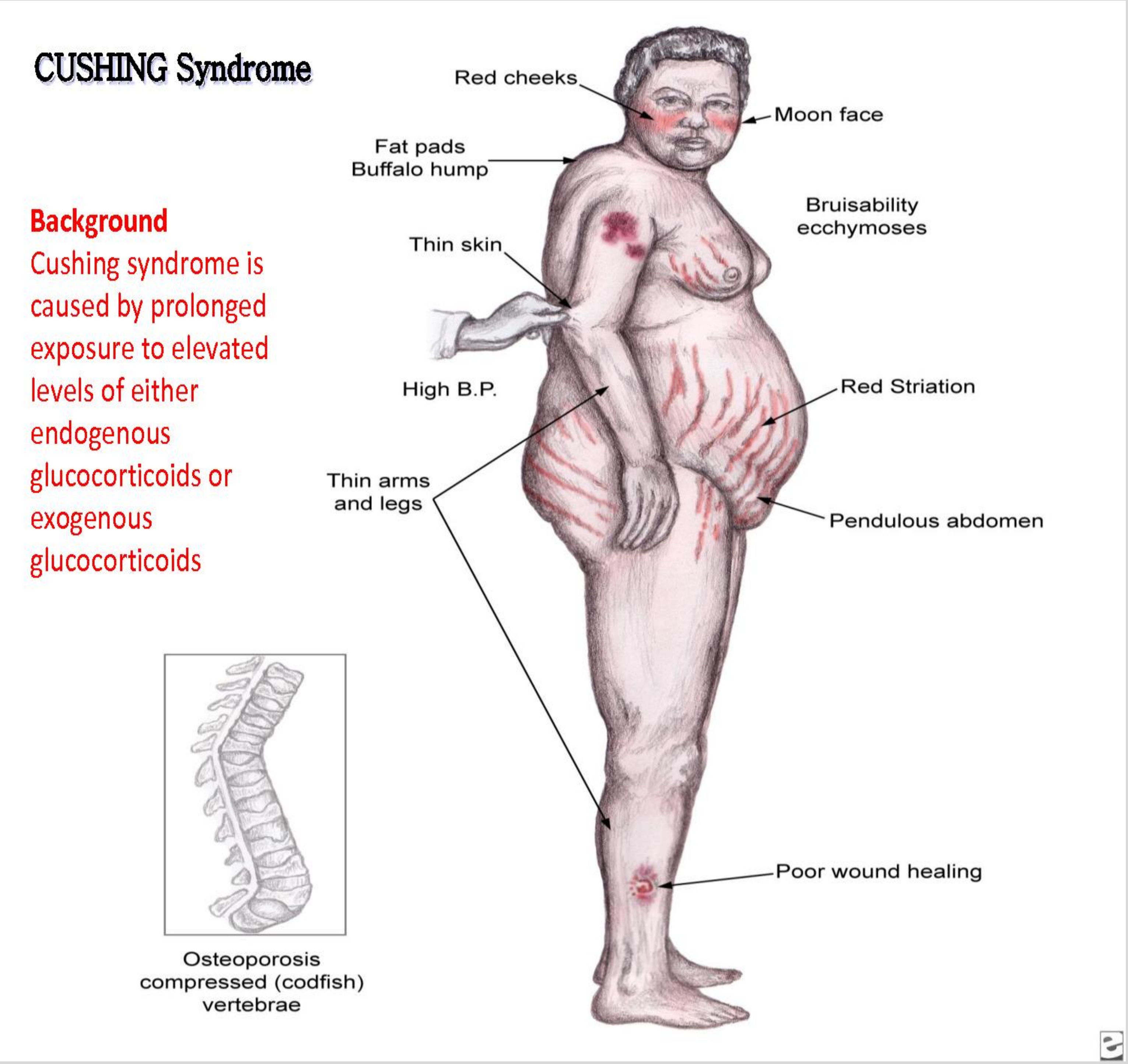
Adverse effects associated with glucocorticoid therapy are dose- and duration-dependent.

The following are typical toxicities of prolonged therapy:

- Hypertension
- Hyperglycemia; polyuria, polydipsia
- Central obesity; buffalo hump; moon facies
- Facial erythema
- Skeletal muscle wasting
- Thin fragile skin
- Menstrual irregularities; hirsuitism; acne; ↓ libido; impotence; infertility
- Cataracts; ↑ocular pressure
- Growth retardation in children
- Osteoporosis; fractures; aseptic necrosis of the femoral head
- Euphoria/depression; psychosis; headache
- ↑ risk of infection; impaired wound healing
- Peptic ulcers

Greater cumulative dose › more likely to cause chronic AEs

Cushingoid effects



Glucocorticoids have numerous effects and act on most body tissues.

These actions on body tissues are demonstrated by the clinically observed effects.

http://www.docstoc.com/docs/77842308/cushing-Index-of

Features of Cushing's Syndrome

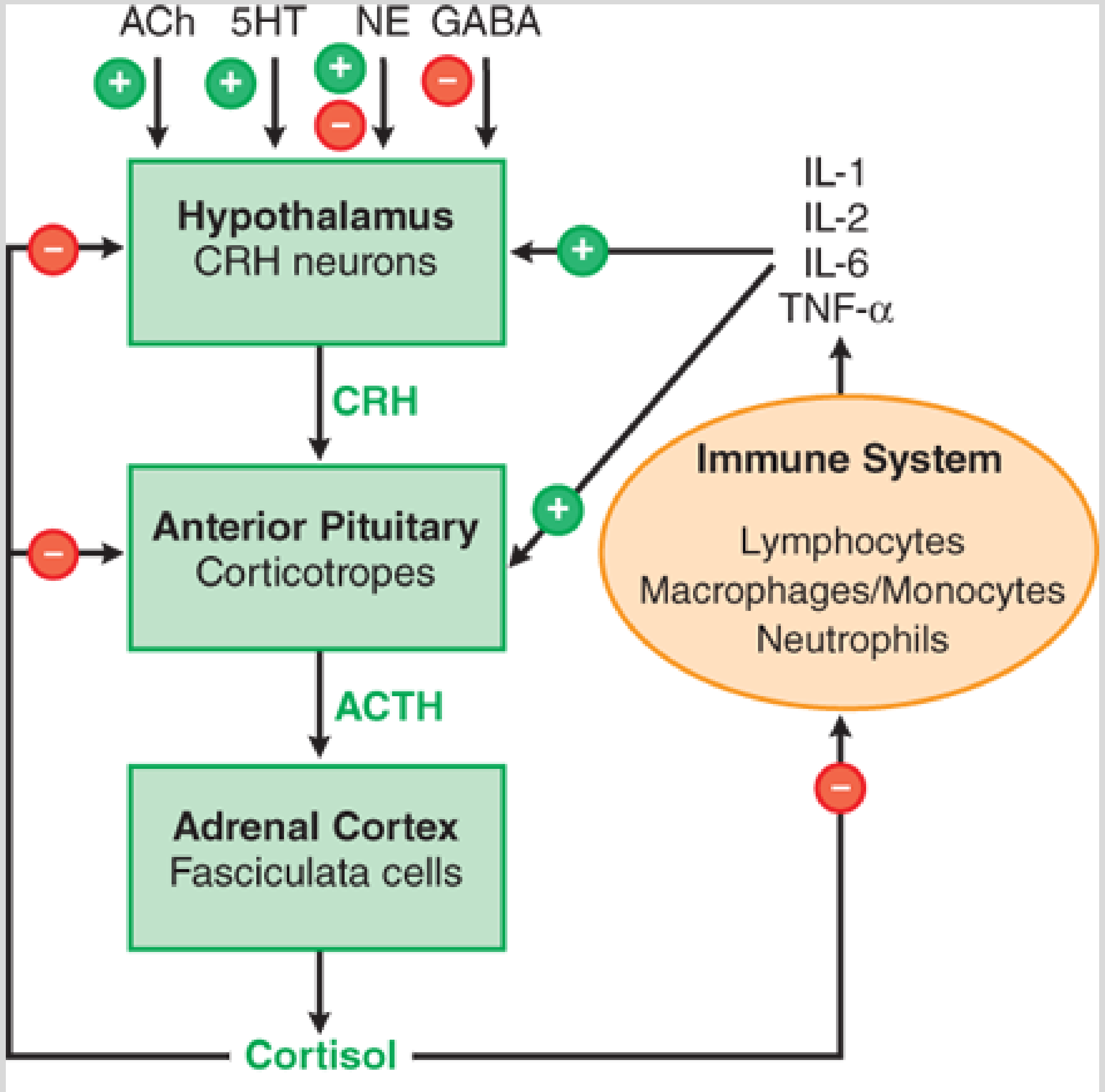
A

Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com

C

Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com

Panel A. Central obesity and broad, purple stretch marks
Panel C. Thin and brittle skin in an elderly patient with Cushing's



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition:

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Complications of abrupt glucocorticoid withdrawal

- **Acute adrenal insufficiency**
  - severe abdominal pains, vomiting, diarrhea
  - profound muscle weakness and fatigue
  - hypotension, hyponatremia, hyperkalemia
  - Shock (adrenal crisis) ⇒ death

➢ Gradually taper the dose of the glucocorticoid drug to prevent adrenal insufficiency.

- **Disease flare**
  - Worsening or new symptoms
  - Increased disease activity

Mechanism of acute adrenal insufficiency:

Glucocorticoid drug therapy suppresses the hypothalamic-pituitary-adrenal (HPA) axis by feedback inhibition → Production of CRH and ACTH are decreased → Production of cortisol in the adrenal cortex decreases → The adrenal glands atrophy. → The adrenal glands are unable to quickly produce cortisol when the glucocorticoid drug is abruptly stopped.

CUSHING'S SYNDROME MNEMONIC

C	Central obesity, Cervical fat pads, Collagen fiber weakness, Comedones (acne)
U	Urinary free cortisol and glucose increase
S	Straie, Suppressed immunity
H	Hypercortisolism, Hypertension, Hyperglycemia, Hypercholesterolemia, Hirsutism, Hypernatremia, Hypokalemia
I	Iatrogenic (Influence of corticosteroids administration)
N	Noniatrogenic (Neoplasms – nonmelanoma skin cancer)
G	Glucose intolerance, Growth retardation

Wikipedia: Cushing's Syndrome

Check your knowledge

What is the goal of therapy in RA?

What is the role of NSAIDs in RA management?

What is the role of prednisone in RA management?

Mary indicated that she doesn't want to take the prednisone, "because it will make my face and belly fat and all those other effects you mentioned." What are the effects Mary is describing?

DMARDs

drugs that slow the radiographic progression of rheumatoid arthritis

Characteristics of DMARDS

Minimal **direct** nonspecific anti-inflammatory or analgesic effects

- NSAIDs must be continued during DMARD therapy.

Clinical benefit is usually delayed for weeks or months.

- Up to 2/3 of patients develop some clinical improvement with DMARD Tx.

Frequently induces an improvement in serologic evidence of disease activity

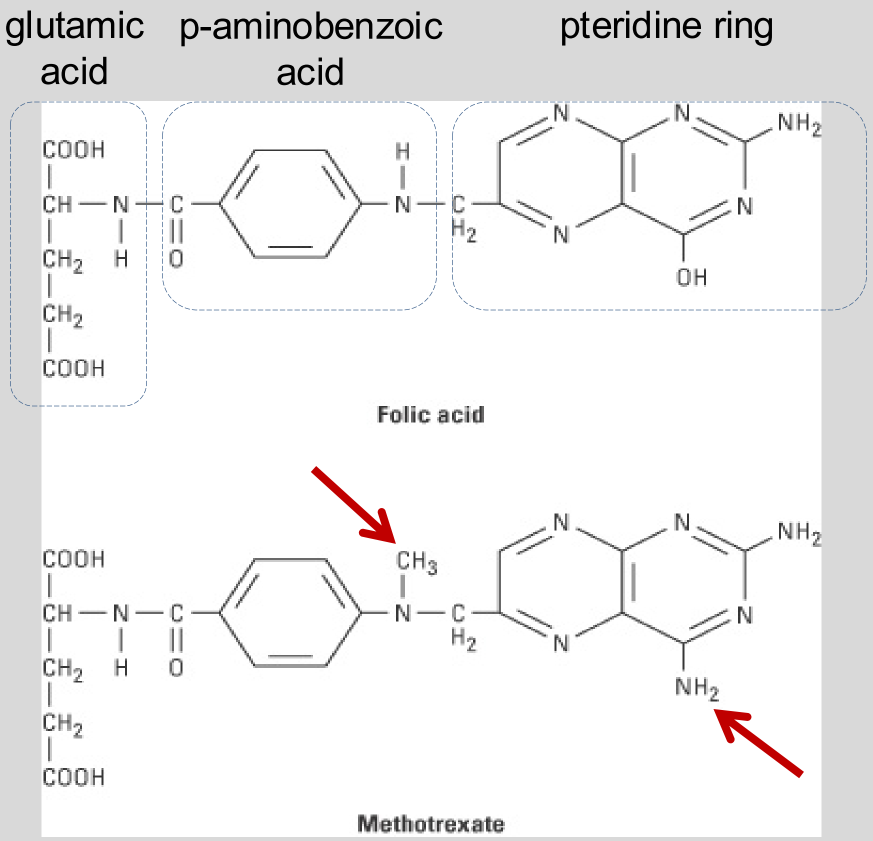
- Titers of rheumatoid factor, C-reactive protein, ESR frequently decline

DMARD therapy, ***especially early in the disease course***, retards the development of bone erosions.

✳Methotrexate (MTX):

a folate analog

an antimetabolite

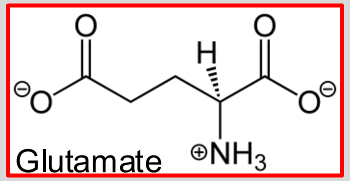


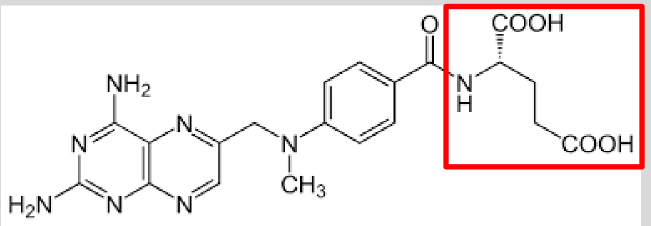
structure-activity relationship

MTX Pharmacokinetic Properties

A	For RA: Oral, subcutaneous, intramuscular Oral bioavailability highly variable, dose dependent, and saturable ➢ Bioavailability decreases as dose increases
D	Protein binding ~50% (narrow therapeutic window) Penetrates body fluids (e.g. pleural effusions, ascites) Sustained concentrations in liver and kidney Does not penetrate BBB OAT, OATP, P-glycoprotein substrate
M	Taken up by proliferating cells and polyglutamated (active, potent) Slow elimination from cells (weeks) (intestinal flora partially metabolize unabsorbed MTX in gut)
E	Renal excretion (80-90% unchanged drug)
t½	Low-dose, adults: 3-10 hours (dose-dependent)

Onset, antirheumatic: 3-6 weeks





MTX Antiproliferative / Anti-inflammatory Effects

1. DHFR inhibition→→→ blocks DNA /RNA synthesis & proliferation → impairs lymphocyte and macrophage function and proliferation
2. TYMS inhibition→→→blocks formation of thymidylate → impairs DNA synthesis and repair → antiproliferative effects
3. AICAR formyl transferase inhibition → AICAR accumulation → inhibits adenosine deaminase and AMP deaminase → extracellular adenosine accumulation → anti-inflammatory effect

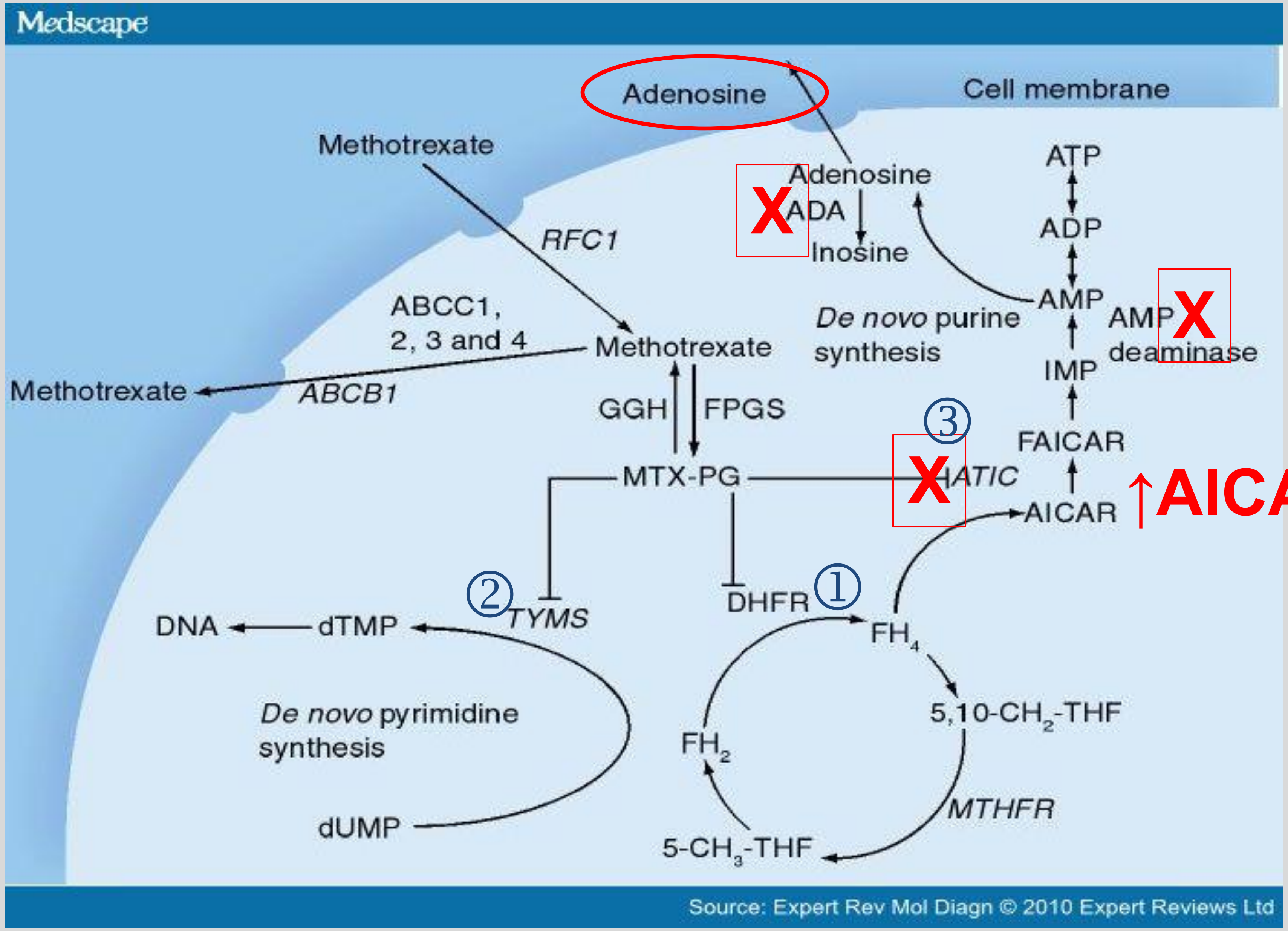
DHFR: dihydrofolate reductase
TYMS: thymidylate synthase
AICAR: 5-aminoimidazole-4-carboxamide ribonucleotide
ATIC: AICAR formyltransferase/IMP cyclohydrolase
FAICAR: 5-formamidoimidazole-4-carboxamide ribotide

Abbreviations key for the following slides

RFC1: reduced folate carrier 1
ABCB1: ABC (efflux pump) B1
FPGS: folylpolyglutamate synthase
GGH: gamma-glutamyl hydrolase
DHFR: dihydrofolate reductase
FH2: dihydrofolate
FH4: tetrahydrofolate
MTHFR: methylenetetrahydrofolate-reductase
TYMS: thymidylate synthase
AICAR:5-amido-imidazole-4-carbox-amide ribo-nucleotide
ATIC: AICAR transformylase

dUMP: deoxyuridine monophosphate
dTMP: deoxythymidine monophosphate (thymidylate)
IMP: inosine monophosphate
AMP: adenosine monophosphate
ADP: adenosine diphosphate
ATP: adenosine triphosphate
ADA: adenosine deaminase

MTX proposed mechanism of anti-inflammatory action



↑AICAR

MTX enters the cell through active transport by reduced folate carrier. Inside the cell, MTX is polyglutamated and can have up to seven glutamic acid moieties (MTXPG2-7)

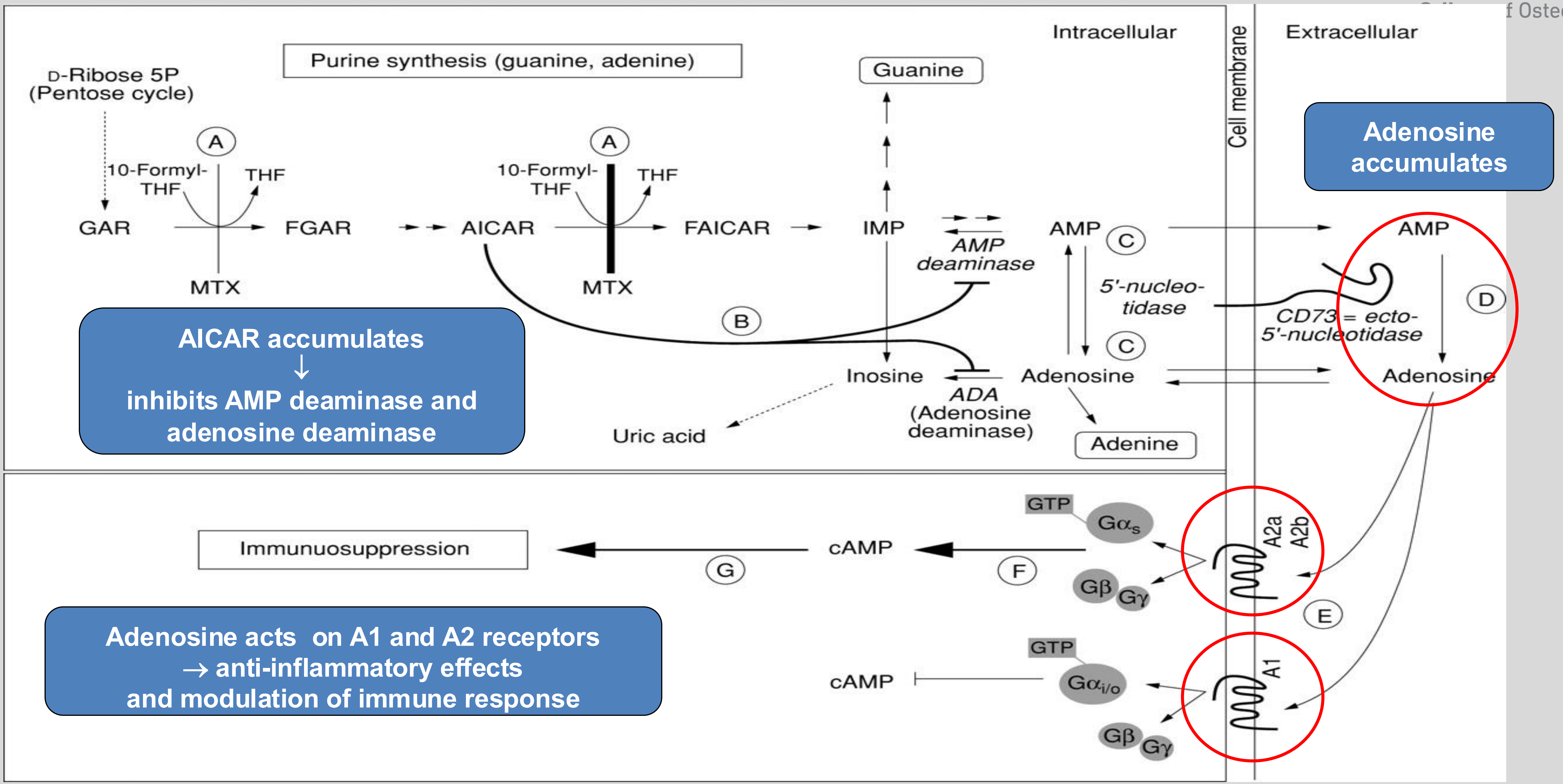
MTX inhibits:
①DHFR,
②TYMS, and
③AICAR transformylase (ATIC)

Accumulation of AICAR and its metabolites inhibits adenosine deaminase and AMP deaminase, which causes intracellular accumulation of adenosine and adenine nucleotides.

antiproliferative
and
anti-inflammatory effects

Zsolt B Nagy; Mónika Csanád; Katalin Tóth; Balázs Börzsönyi; Csaba Demendi; János Rigó Jr; József Gábor Joó. Current Concepts in the Genetic Diagnostics of Rheumatoid Arthritis. Expert Rev Mol Diagn. 2010;10(5):603-618. © 2010 Expert Reviews Ltd.

MTX Anti-Inflammatory Effects in RA (probable mechanism)



M Cutolo, A Sulli, C Pizzorni, B Seriolo. Anti-inflammatory mechanisms of methotrexate in rheumatoid arthritis. *Ann Rheum Dis* 2001;60:729-735 doi:10.1136/ard.60.8.729 http://ard.bmj.com/content/60/8/729.full

MTX Effects

Anti-inflammatory effects:

MTX suppresses inflammatory functions of:

- Neutrophils
- Macrophages
- Dendritic cells
- Lymphocytes
- PMN chemotaxis (secondary effects)

Antiproliferative effects:

- Direct inhibitory effects on proliferation
- Stimulates apoptosis
- Inhibits proinflammatory cytokines linked to rheumatoid synovitis

Some effect on DHFR, which affects lymphocyte and macrophage function, but this is not its principal mechanism of antirheumatic action.

MTX is the **First-line** DMARD and anchor in combination therapies

- Oral once-weekly dosing
- Proven clinical benefits

  improved survival – CV and all-cause mortality
- Long-term sustained efficacy
- Well-understood toxicity profile
- Combination with other RA agents
- Approved in idiopathic juvenile arthritis
- (and many uses in autoimmune disorders and in oncology/hematology)

MTX Adverse Effects / Cautions

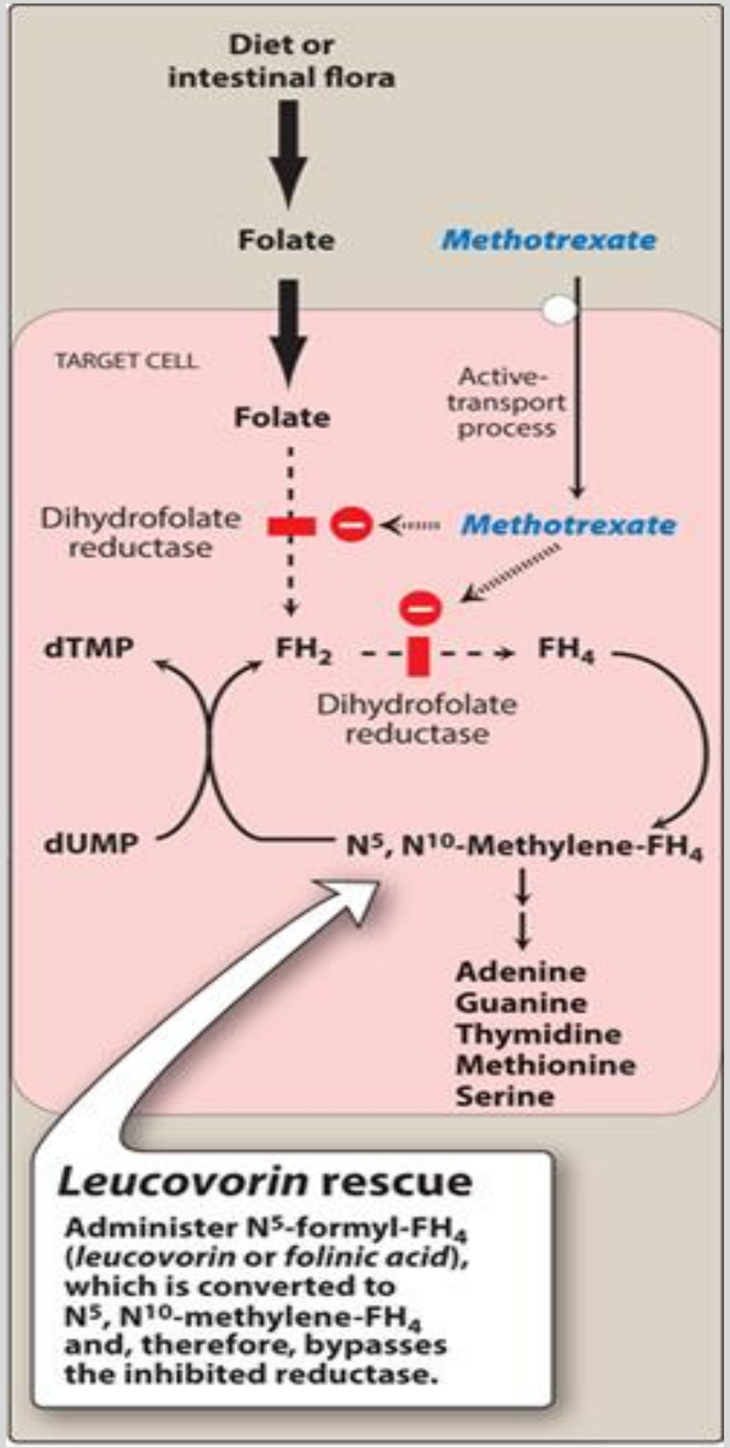
- Gastrointestinal toxicity common
- Hepatotoxicity

ranges from minor increase in aminotransferases to cirrhosis and liver failure

- Bone marrow toxicity

All patients:
Folic acid daily supplementation to reduce GI, bone marrow suppression and liver function test abnormalities.

- Nephrotoxicity
- Lymphomas
- Pneumonitis (rare but life-threatening)
- Dermatologic, Alopecia
- Phototoxicity
- Opportunistic infections
- Infertility / menstrual irregularities
- **Pregnancy / breast feeding contraindicated**
- Drug interactions



Folic acid supplementation

Folic acid (vitamin B9) can help replete folate stores

in patients who take MTX chronically.

If inadequate response to folic acid, switch to leucovorin (folinic acid) once weekly 8-12 h after MTX dose.

Leucovorin is a reduced form of folate.

It bypasses DHFR inhibited by MTX and repletes stores of reduced folate.

Lippincott Illustrated Reviews Pharmacology Figure 46.9

Methotrexate: Pharmacokinetic mechanisms of drug interactions

(MTX is not metabolized in liver or kidney)

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Transport:	Plasma protein binding:	Mechanism-related
Inhibition of OAT and P-gp → ↑MTX concentration → ↑MTX toxicity risk OAT-mediated: – NSAIDs, aspirin, probenecid, penicillins, sulfonamides, many others P-glycoprotein-mediated: – cyclosporine, many others OAT: organic anion transporter P-gp: P-glycoprotein	Displacement of MTX → ↑ free MTX → ↑risk MTX toxicity – Salicylates – Phenytoin – Sulfonamides – Doxycycline – others	Shared toxicities • leflunomide (hepatotoxicity; myelosuppression; others) • azathioprine (hepatotoxicity) Immunosuppessants • Increase infection risk Vaccines – live attenuated • decreased immunogenicity • potential risk of infection from the vaccine antigen Administration of live attenuated virus and bacterial vaccines during immunosuppressive therapy is a contraindication of all immunosuppressants.

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Check your knowledge

Mary was initiated on the initial DMARD of choice. What was the drug and what was the dosing schedule?

Mary is given information on the potential side effects of the drug, including the signs and symptoms of serious adverse effects. What are they?

Mary was advised that daily supplementation with a particular vitamin is important for reducing some of the side effects of the drug. What is the vitamin and how does it help?

Check your knowledge

Why is it important that Mary was asked about contraception and plans for becoming pregnant?

What are the mechanisms of methotrexate's immune modulating and anti-inflammatory effects?

Summary of DMARDS Part 1
Glucocorticoids and Methotrexate

- The disease-modifying antirheumatic drugs (DMARDS) are a heterogenous group of drugs that have anti-inflammatory actions and are recommended early in the RA to slow the progression of joint damage by modifying the underlying disease. It may take several weeks to months for the antirheumatic effects to become apparent.
- NSAIDs are commonly used for symptomatic relief but they do not alter the disease progression.
- Glucocorticoids promptly and dramatically bring the level of inflammation under control and they are capable of slowing the appearance of new bone erosions. Prolonged use of glucocorticoids causes adrenal suppression and also leads to serious and disabling toxic effects.

- The DMARDS are sub-categorized into the conventional synthetic, biological, and targeted small-molecule synthetic drugs.
- Methotrexate is the first-line DMARD and “anchor” in combination therapies for RA in patients with moderately to severely active disease or for those failing to respond to a single agent. Combinations of nonbiological DMARDS or a biological DMARD with methotrexate have shown improved efficacy.
- MTX side effects include nausea and mucosal ulcers (most common toxicities), leukopenia, anemia, stomatitis, GI ulcerations, and alopecia, which are probably the result of inhibiting cellular proliferation. Progressive dose-related hepatotoxicity in the form of enzyme elevation occurs frequently.
- Daily folic acid or weekly leucovorin may reduce the incidence of GI and liver function test abnormalities.

Check your knowledge: Answers

What is the goal of therapy in RA?

- To achieve and maintain remission or a state of low disease activity – without compromising safety.

What is the role of NSAIDs in RA management?

- To control inflammation and provide analgesia – full anti-inflammatory doses should be used initially for active RA – unless contraindicated (of course!) by gastrointestinal, cardiovascular, or kidney disease.
- For example, ibuprofen 400 mg (2 tabs) every four to six hours.

What is the role of prednisone in RA management?

- To control inflammation. Prednisone and other glucocorticoids rapidly reduce inflammatory synovitis in patients with RA.

Mary indicated that she doesn't want to take the prednisone, "because it will make my face and belly fat and all those other effects you mentioned." What are the effects Mary is describing?

- Glucocorticoids mimic the effects of cortisol. High anti-inflammatory doses cause fat redistribution (moon facies, nuchal fat, abdominal fat), increased appetite, and water retention.
- See slide 34 for the entire list. The mechanisms are described in the Notes packet.

Check your knowledge: Answers

Mary was initiated on the initial DMARD of choice. What was the drug and what was the dosing schedule?

- Oral (usually), low-dose, once-weekly MTX has **strong evidence** of effectiveness, sustained response, manageable toxicities, and improved quality of life in the treatment of RA and several other forms of inflammatory arthritis and autoimmune diseases.
- MTX is referred to as “the anchor”. It has become the cornerstone of RA management. It can be used as monotherapy or combined with many other DMARDs.

Mary is given information on the potential side effects of the drug, including the signs and symptoms of serious adverse effects. What are they?

- Low-dose once weekly MTX is usually well-tolerated. Common side effects are stomatitis and GI upset.
- Patients should be closely monitored for bone marrow, liver, and lung toxicity.

Mary was advised that daily supplementation with a particular vitamin is important for reducing some of the side effects of the drug. What is the vitamin and how does it help?

- All patients on weekly MTX should take daily folic acid supplementation. Folic acid can reduce stomatitis (mouth sores), mucositis (which causes GI upset, diarrhea) and bone marrow and liver function test abnormalities.
- Folic acid daily supplementation does not reduce the efficacy of MTX.

Check your knowledge: Answers

Why is it important that Mary was asked about contraception and plans for becoming pregnant?

- Methotrexate is fetotoxic – spontaneous abortion, skull anomalies, facial dysmorphism, CNS, limb, and cardiac abnormalities, intellectual impairment. Exposure in the second and third trimesters are associated with intrauterine growth restriction and functional abnormalities.

What are the mechanisms of methotrexate's immune modulating and anti-inflammatory effects?

- DHFR: MTX inhibits dihydrofolate reductase, which inhibits folate activity, which inhibits purine synthesis.
- TYMS: MTX inhibits thymidylate synthase necessary for DNA synthesis.
- Adenosine: MTX inhibits AICAR transformylase, which leads to accumulation of AICAR, which inhibits adenosine deaminase and AMP deaminase, which increases extracellular concentrations of adenosine, which acts on it's A2 and A1 receptors, which activates immunomodulatory and anti-inflammatory effects.

THE ART OF MEDICINE

“Doctor and Doll”
Norman Rockwell (1929)

References

- Access Medicine Current Diagnosis & Treatment: Rheumatology, 4e, 2021; Chapter 13: Rheumatoid Arthritis
- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics 14e, 2023; Chapter 50: ACTH, Adrenal Steroids, and Pharmacology of the Adrenal Cortex; Chapter 70: Cytotoxics and Antimetabolites; Chapter 71: Protein Kinase and Pathway-Targeted Small Molecules
- Access Medicine Harrison's Principals of Internal Medicine, 21e, 2022; Chapter 358: Rheumatoid Arthritis
- Access Medicine Katzung's Basic & Clinical Pharmacology 16e, 2024; Chapter 36: NSAIDS, DMARDS, Nonopioid Analgesics, and Drugs Used in Gout; Chapter 39: Adrenocortical Steroids and Adrenocortical Antagonists
- Clinical Key Hochberg, et.al. Rheumatology, 7e, 2019; Chapter 59: Drugs and Pregnancy; Chapter 64: Systemic glucocorticoids in rheumatology
- Lexicomp Feb 2024
- UpToDate: various topics; literature reviews current

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